MATH 121: COMMUTATIVE ALGEBRA

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1. Rings

Important Example: Let X be a "geometric set", and consider the ring

$$\mathrm{Hom}(X, \mathbb{R}) = \{X \xrightarrow{f} \mathbb{R}\}$$

with pointwise addition and multiplication

$$(f+g)(x) = f(x) + g(x), \quad (fg)(x) = f(x)g(x)$$

$r \in \mathrm{Hom}(X, \mathbb{R})$ is a unit if and only if there exists $g \in \mathrm{Hom}(X, \mathbb{R})$ such that $fg = 1$ if and only if $\forall x \in X, (fg)(x) = f(x)g(x) = 1$

$$\mathrm{Hom}(X, \mathbb{R})^\times = \mathrm{Hom}(X, \mathbb{R}^\times)$$

Definition 1.1. *Let $a \in R$, $a \neq 0$*

- *a is a unit if and only if $\exists b \in R$ such that $ab = 1$,*
- *a is a zero divisor if and only if $\exists b \in R \setminus \{0\}$ such that $ab = 0$,*

Lemma 1.2. *The set of units $R^\times = \{a \in R \,|\, a \text{ is a unit}\}$ is a group under multiplication.*

Proof: Worksheet (03/31).

Lemma 1.3 (Useful Lemma). *If $\phi : R \to S$ is a ring homomorphism, then it follows from the definition that*

- $\phi(0_R) = 0_S$
- $\phi(-a) = -\phi(a)$
- $\phi(a^{-1}) = \phi(a)^{-1}$

Remark 1.4. *$\mathbb{Z}$ is initial in the category of rings. There is a unique homomorphism from the initial object to every other object. For any ring R,*

$$Hom(\mathbb{Z}, R) = \{1 \to 1_R\}$$

We don't know anything about homomorphisms to the initial obejct.

2. Ideals

Lemma 2.1. *If $\phi : R \to S$ is any ring homomorphism and $J \subset S$ is an ideal $\phi^{-1}(J)$ is also an ideal.*

Proof: Worksheet Q4.1.

Theorem 2.2. *If $I \subset R$ is an ideal then,*

(1) There exists a ring R/I and a surjective ring homomorphism $\pi : R \to R/I$ such that $\ker(\pi) = I$*. Moreover R/I and π are unique upto isomorphism.*
(2) There is an inclusion preserving bijection

$$\{\text{Ideals in } R/I\} \leftrightarrow \{\text{Ideals in } R \text{ containing } I\}$$
$$\overline{J} \leftrightarrow \pi^{-1}(\overline{J})$$

Proof.

(1) Construct R/I explicitly.
(2) I gives an equivalence relation $\sim$ on R such that

$$a \sim b \iff a - b \in I \iff a = b + I$$

Show that equivalence classes $R/\sim$ is a ring.

$$R \to R/I = R/\sim$$
$$a \mapsto \overline{a} = a + I$$

Let $\overline{J}$ be an ideal in R/I, and J be an ideal in R and define the maps

$$\phi : \overline{J} \to \pi^{-1}(\overline{J}), \quad \psi : J \to \pi(J)$$

Now we want to show that $\phi \circ \psi = \psi \circ \phi = id$. Consider $\phi \circ \psi(J)$ and $\psi \circ \phi(\overline{J})$. Note that

$$\psi \circ \phi(\overline{J}) = \pi(\pi^{-1}(\overline{J})) = \overline{J}$$

since π is surjective. Thus $\psi \circ \phi = id$.

Now, note that $J \subseteq \phi \circ \psi(J) = \pi^{-1}(\pi(J))$, and since $\overline{0} \in \pi(J)$, $I \subseteq \pi^{-1}(\pi(J))$. An element $x \in \pi^{-1}(\overline{J})$ looks like $x + I$, and $\pi^{-1}(\overline{J}) = J + I$ Moreover since $\overline{0} \in \overline{J}$, we have $\pi^{-1}(\overline{0}) \subseteq \pi^{-1}(\overline{J})$ which implies $I \subseteq J$. Thus, $\psi \circ \phi(J) = J$ and $\psi \circ \phi = id$.

□

Exercise: Describe the ideals in

(1) $\mathbb{Z}/\langle 24\rangle = \{\langle 0\rangle, \langle 1\rangle, \langle 2\rangle, \langle 3\rangle, \langle 4\rangle, \langle 6\rangle, \langle 8\rangle, \langle 12\rangle\}$
(2) $\mathbb{Z}/\langle n\rangle = \{\langle d\rangle : d|n\}$
(3) $\mathbb{Z}/\langle 3\rangle = \{\langle 0\rangle, \langle 1\rangle\}$
(4) $\mathbb{Z}/\langle 5\rangle = \{\langle 0\rangle, \langle 1\rangle\}$
(5) $\mathbb{C}[x,y]/\langle y - x^2\rangle \cong \mathbb{C}[x] = \{\text{Ideals in } \mathbb{C}[x]\}$
(6) $\mathbb{C}[x]/\langle x^n\rangle = \{\langle x^i\rangle : 0 \le i < n\}$

Lemma 2.3. *If $\phi : R \to S$ is a ring homomorphism with* $\ker(\phi) \supseteq I$ *then there exists a unique morphism $\widetilde{\phi} : R/I \to S$*

$$\begin{tikzcd} R \arrow[r, "\pi"] \arrow[d, "\phi"'] & R/I \arrow[dl, dashed, "\widetilde{\phi}"] \\ S & \end{tikzcd}$$

Universal Property of Quotinents: *Upto isomorphism R/I is the ring such that*

$$\operatorname{Hom}(R/I, S) = \{\phi : R \to S : I \subset \ker(\phi)\}$$

We want to define $\widetilde{\phi}(\overline{a}) = \widetilde{\phi}(a+I) = \phi(a)$ and $\widetilde{\phi}(\overline{0}) = 0$.

Lemmact 2.4. *Maximal Ideal $\implies$ Prime Ideals.*

Definition 2.5. *The spectrum of a ring R is the set*

$$\operatorname{Spec}(R) = \{\textit{Prime Ideals in } R\}$$

the maximal spectrum of R is the set

$$m\operatorname{Spec}(R) = \{\textit{Maximal Ideals in } R\}$$

$\operatorname{Spec}(R)$ *is where the 'geometry lives'. Note that $m\operatorname{Spec}(R) \subset \operatorname{Spec}(R)$.*

Example: *$R = \mathbb{Z}[\sqrt{-5}]$. 2 is irreducible but not prime. $R/\langle 2\rangle = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.*

Example: *$R = \mathbb{C}[x,y]/\langle x^3 - y^2\rangle$. x is irreducible, but $\langle x\rangle$ is not prime. $R/\langle x\rangle = \mathbb{C}[y]/\langle y\rangle$ and y is a zero divisor.*

Theorem 2.6. *Let R be a non-zero ring.*

(1) R has a maximal ideal.
(2) If $I \subsetneq R$ is a proper ideal there exists a prime ideal $P \subseteq R$ such that $I \subset P \subsetneq R$.

Proof. **Zorn's Lemma:** Let $(P, \leq)$ be a po-set.

If $P \neq \emptyset$ and if $C = C_1 \leq C_2 \leq C_3 \leq \cdots$ is a chain in $(P, \leq)$ then there exists $B \in P$ such that $C_i \leq B$ for all i, then there exists $M \in P$ such that if $S \leq M$ then $M = S$.

Let $P = \{\text{Proper ideals in } R\}$ and $\leq$ be inclusion. We claim that $(P, \leq)$ is a poset.

(1)
- $P \neq \emptyset$ since $\langle 0\rangle \in P$.
- Given a chain in P

$$I_1 \subseteq I_2 \subseteq \cdots$$

Then, define $B = \bigcup I_j$. Note that B is an ideals since union of nested ideals is an ideal. And B is proper since $1 \notin I_j$ for all j, and hence $1 \notin B$. Then there exists some ideal $M \subset R$ such that M is maximal.

(2) Follows from (1). Let $R = R/I$. Pre-image of prime ideals is prime.

Thus, $m\operatorname{Spec}(R) \neq \emptyset$ and $\operatorname{Spec}(R) \neq \emptyset$. ☐

3. Zariski Topology

Definition 3.1. *If $S \subseteq R$ is any subset of R,*

$$\mathbb{V}(S) = \{[P] \in \operatorname{Spec}(R) | S \subseteq P\} \subseteq \operatorname{Spec}(R)$$

Definition 3.2. *An algebraic subset of $\mathbb{K}^n$ of some field K is a set of the form*

$$\mathbb{V}(\mathcal{S}) = \{P \in \mathbb{K}^n | f(p) = 0, \forall f \in \mathcal{S}\}$$

for a subset $\mathcal{S} \subset \mathbb{K}[x_1, \cdots, x_n]$

Examples:

- $\mathbb{V}(x^2+y^2-1) \subseteq \mathbb{R}^2$
$$\{(x,y) \in \mathbb{R}^2 | x^2+y^2-1=0\}$$
- $\mathbb{V}(x^2+y^2-1) \subseteq \mathbb{R}^3$
$$\{(x,y,z) \in \mathbb{R}^3 | x^2+y^2-1=0\}$$
- $\mathbb{V}(x^2-y) \subseteq \mathbb{R}[x,y]$
$$\{(x,y) \in \mathbb{R}[x,y] | x^2=y\}$$
- $\mathbb{V}(x^2-xy) \subseteq \mathbb{R}[x,y]$
$$\{(x,y) \in \mathbb{R}[x,y] \,|\, x=0,\ x-y=0\}$$
- $\mathbb{V}(x^2+1) \subseteq \mathbb{R}[x]$
$$\mathbb{V}(x^2+1)=\emptyset$$
- $\mathbb{V}(x^2+1) \subseteq \mathbb{C}[x]$
$$\mathbb{V}(x^2+1)=\{i,-i\}$$
- $\mathbb{V}(x^2+4) \subseteq \mathbb{F}_3[x]$
$$\mathbb{V}(x^2+4)=\emptyset$$
- $\mathbb{V}(x^2+4) \subseteq \mathbb{F}_p[x]$
TODO:
- $\mathbb{V}(x^6-3x^5y+3x^4y^2-x^3y^3) \subseteq \mathbb{R}[x,y]$

$$\mathbb{V}(x^6-3x^5y+3x^4y^2-x^3y^3)=\mathbb{V}(x^3(x-y)^3)=\{(x,y) \in \mathbb{R}[x,y] \,|\, x=0,\ x-y=0\}$$

- $\mathbb{V}(n(x^2+y^2+z^2), n \in \mathbb{Z}) \subseteq \mathbb{R}^3$
TODO:
- $\mathbb{V}$(a collection of any linear polynomials in $\mathbb{K}[x_1,\cdots,x_n]$)
TODO:

Definition 3.3. *Let $V \subseteq \mathbb{K}^n$ be an algebraic set. The defining ideal of V is*
$$\mathbb{I}(V)=\{f \in \mathbb{K}[x_1,\cdots,x_n] \,|\, f(p)=0, \forall p \in V\}$$

Definition 3.4. *If $\mathfrak{M} \subseteq R$ is a maximial ideal, the residue field of $\mathfrak{M}$ is $R/\mathfrak{M}$.*

Definition 3.5. *Let $\mathbb{K}$ be a field. A $\mathbb{K}$ algebra is the data of a ring R together with a fixed ring homomorphism*
$$\alpha : \mathbb{K} \to R$$

Definition 3.6. *If $(R, \mathbb{K} \xrightarrow{\alpha} R)$ and $(S, \mathbb{K} \xrightarrow{\beta} S)$ are $\mathbb{K}$-algebras, a ring map $\phi : R \to S$ is a morphism of $\mathbb{K}$-algebras iff*

$$\begin{tikzcd} R \arrow[r, "\phi"] & S \\ \mathbb{K} \arrow[u, "\alpha"] \arrow[ur, "\beta"] & \end{tikzcd}$$

Lemmact 3.7. *If R is k-algebra, R/I has a natural k-algebra structure compatible with the quotinent map.*

$$\begin{tikzcd} R \arrow[r, "\pi"] & R/I \\ k \arrow[u, "\alpha"] \arrow[ur, dashed, "\exists!"'] & \end{tikzcd}$$

Definition 3.8. *A* valuation ring *of a field $\mathbb{K}$ is a subring $R \subset \mathbb{K}$ with the property that for all $x \in \mathbb{K}^\times$ either $x \in R$ or $x^{-1} \in R$.*

4. Gröbner basis

Remark 4.1. *Ideal Membership Problem: Suppose $I \subseteq R[x_1, \cdots, x_n]$ an ideal. How can we algorithmcally determine if $f \in I$?*

Definition 4.2. *A ring R is Noetherian if and only if every ascending chain of ideals stabilizes (Ascending Chain Condition), i.e.*

$$I_1 \subseteq I_2 \subseteq \cdots$$

there exists $N \in \mathbb{Z}$ such that $I_k = I_{k+1}$ for all $k \geq N$.

Lemma 4.3. *R is Noetherian if and only if every ideal is finitely generated.*

Lemma 4.4 (Dickson's Lemma)**.** *Every monomial ideal in $\mathbb{K}[x_1, \cdots, x_n]$ is finitely generated.*

Theorem 4.5 (Hilbert's Basis Theorem)**.** *If R is a Noetherian ring, then $R[x]$ is also Noetherian.*

5. Nullstellensatz

Theorem 5.1 (Hilbert's Nullstellensatz I)**.** *If $\mathbb{K}$ is algebraically closed, then every maximal ideal of $\mathbb{K}[x_1, \ldots, x_n]$ is of the form $\langle x_1 - a_1, \ldots, x_n - a_n \rangle$ for some point $(a_1, \ldots, a_n) \in \mathbb{K}^n$.*

Theorem 5.2 (Hilbert's Nullstellensatz II)**.** *If $\mathbb{K}$ is algebraically closed and $V = \mathbb{V}(\mathcal{S}) \subset \mathbb{K}^n$ is an algebraic set, for some subset $\mathcal{S} \subset \mathbb{K}[x_1, \ldots, x_n]$, then $\mathbb{I}(V) = \sqrt{\langle \mathcal{S} \rangle}$.*

Theorem 5.3 (Hilbert's Weak Nullstellensatz)**.** *If $\mathbb{K}$ is algebraically closed and $I \subset \mathbb{K}[x_1, \ldots, x_n]$ is an ideal then $\mathbb{V}(I) = \varnothing$ if and only if $I = \langle 1 \rangle$.*

Theorem 5.4 (Hilbert's Nullstellensatz III)**.** *Let $\mathbb{K}$ be an algebraically closed field. If $V \subset \mathbb{K}^n$ is an algebraic subset then the maps $\mathbb{I}_V$ and $\mathbb{V}_V$ induce inverse, inclusion-reversing bijections:*

$$\begin{tikzcd} \left\{ \begin{array}{c} \textit{algebraic subsets} \\ \textit{in } V \end{array} \right\} \arrow[r, bend left, "\mathbb{I}_V"] & \left\{ \begin{array}{c} \textit{radical ideals} \\ \textit{in } \mathbb{K}[V] \end{array} \right\} \arrow[l, bend left, "\mathbb{V}_V"] \end{tikzcd}$$

Remark 5.5. $\mathbb{I}$ *and* $\mathbb{V}$ *properties:*

(1) *If* $\mathcal{S} \subset \mathcal{T}$ *then* $\mathbb{V}(\mathcal{T}) \subset \mathbb{V}(\mathcal{S})$
(2)
$$\mathbb{V}(\mathcal{S}) = \mathbb{V}(\langle \mathcal{S} \rangle) = \mathbb{V}(\sqrt{\langle \mathcal{S} \rangle})$$
(3) $\mathbb{I}(V)$ *is radical.*
(4) $\langle \mathcal{S} \rangle \subset \sqrt{\langle \mathcal{S} \rangle} \subset \mathbb{I}(\mathbb{V}(\mathcal{S}))$
(5) $\mathbb{V}(\mathcal{S}) = \mathbb{V}(\mathbb{I}(\mathbb{V}(\mathcal{S})))$
(6) *If* $fg \in \mathbb{I}(V)$ *then* $V \subseteq \mathbb{V}(g) \cap \mathbb{V}(f)$.

6. Localization

Definition 6.1. *A subset* $U \subset R$ *is* ***multiplicatively closed*** *if and only if*

(1) $1_R \in U$
(2) *and if* $s, u \in U$ *then* $su \in U$.

Example 6.1.

(1) $R = \mathbb{Z}$, $U = \{1, 3, 9, \ldots\}$ *is multiplicatively closed.*
(2) $R = \mathbb{Z}$, $U = \{1, 3, 9\}$ *is not multiplicatively closed.*
(3) $R = \mathbb{R}[x, y]$, $U = \{f \in R \mid f(1, 0) \neq 0\}$ *is multiplicatively closed.*
(4) $R = \mathbb{R}[x, y]$, $U = \{f \in R \mid f(1, 1) \neq 0,\ or\ f(0, 0) \neq 0\}$ *is not multiplicatively closed.*

Definition 6.2. *Let* R *be a ring and* $U \subseteq R$ *be a multiplicative closed set. The* ***localization*** *of* R *at* U *is a ring* $U^{-1}R$ *together with a ring map* $\eta : R \to U^{-1}R$ *that is universal in the following sense:*

(1) *Given a ring map* $\psi : R \to T$ *such that* $\psi(U) \subseteq T^{\times}$ *there exists a unique ring map* $\widetilde{\psi} : U^{-1}R \to T$ *such that*

$$\begin{tikzcd} R \arrow[r, "\eta"] \arrow[dr, "\psi"'] & U^{-1}R \arrow[d, dashed, "\widetilde{\psi}"] \\ & T \end{tikzcd}$$

Construction of $U^{-1}R$:

Consider $R \times U$ as a set (r_1, u_1) where $r_1 \in R, u_1 \in U$ " $=$" $\frac{r_1}{u_1}$. Define an equivalence relation
$$(r_1, u_1) \sim (r_2, u_2) \iff t(u_2 r_1 - u_1 r_2) = 0, \qquad t \in U$$
Note if R is a domain $t = 1_R$.

Define $\frac{r}{u}$ to be the equivalence class of (r, u) in $R \times U / \sim$.

Set $U^{-1}R = R \times U / \sim = \left\{ \frac{r}{u} \mid r \in R,\ u \in U \right\} / \sim$

Make $U^{-1}R$ a ring with

$$\left(\frac{r_1}{u_1}\right)\left(\frac{r_2}{u_2}\right) = \frac{r_1 r_2}{u_1 u_2}, \qquad \frac{r_1}{u_1} + \frac{r_2}{u_2} = \frac{r_1 u_2 + r_2 u_1}{u_1 u_2}$$

Additive Identity: $0/1$, **Multiplicative Identity:** $1/1$

$$\eta : R \to U^{-1}R$$

Then

$$\begin{aligned}
\ker(\eta) &= \{r \in R \mid \eta(r) = 0\} \\
&= \{r \in R \mid \frac{r}{1} = \frac{0}{1}\} \\
&= \{r \in R \mid t(r \cdot 1 - 1 \cdot 0) = 0,\ t \in U\} \\
&= \{r \in R \mid tr = 0,\ t \in U\}
\end{aligned}$$

If R is a domain, $\ker(\eta) = \langle 0 \rangle \implies \eta$ is injective, unless $0 \in U$.

Example 6.2. ***TODO:*** *Let R be any ring and $U = \{0, 1\}$. Then*

$$U^{-1}R = \left\{\frac{r}{0} \mid r \in R\right\} \cup \left\{\frac{r}{1} \mid r \in R\right\}$$

Let $\frac{r}{1} \stackrel{?}{=} \frac{0}{1} \iff t(r - 0) = 0 \iff tr = 0$ for some $t \in U$. We can take $t = 0$ and make this always true (same argument $\frac{r}{0} = \frac{0}{1}$). Hence $U^{-1}R$ is the zero ring.

Lemma 6.3. *Let $U \subseteq R$ be multiplicatively closed. If U contains a nilpotent then $U^{-1}R$ is the zero ring.*

Definition 6.4. *If $P \subseteq R$ is a prime ideal then $U = R \setminus P$ is multiplicatively closed. Here we write $U^{-1}R = R_P$ and say R **localized at** P.*

Lemma 6.5.

$$PR_p = \left\{\frac{r}{u} \mid r \in P,\ u \in R \setminus P\right\} \subseteq R_P$$

is a maximal ideal.

Proof. Let $PR_P \subsetneq I \subseteq R_p$. Then for $r \in I$ $\frac{r}{u}$ with $r \notin P \implies \frac{r}{u}$ with $r, u \in R \setminus P \implies \frac{u}{r} \in R_p \implies \frac{r}{u}$ is a unit in $R_p \implies I = R_p$. **TODO:** □

Proposition 6.6. *R_P is a local ring (i.e., PR_P is the unique maximal ideal in R_P).*

Proof. If $I \subseteq R_p$ is a maximal ideal, then $I \cap R \setminus P \neq \emptyset$, that is I contains a unit, and hence $I = R_p$. □

Example 6.3. *Let $f \in R$, and $U = \{1, f, f^2, \ldots\}$, then*

$$U^{-1}R = R\left[\frac{1}{f}\right] = \left\{\frac{r}{u} \mid r \in R,\ u \in U\right\} = \left\{\frac{r}{f^k} \mid r \in R,\ k \in \mathbb{Z}_{\geq 0}\right\}$$

that is $U^{-1}R$ is the ring of polynomials in $1/f$ with coefficients in R.

Lemma 6.7.
$$R\left[\frac{1}{f}\right] \cong \frac{R[x]}{\langle xf-1\rangle}$$

Proof. $R[x] \to R\left[\frac{1}{f}\right]$, $x \mapsto \frac{1}{f}$ and $1 \mapsto \frac{1}{1}$. This is surjective because for every $rx^k \in U^{-1}R$ we have $rx^k \mapsto \frac{r}{f^k}$. To show injectivity we must show that $\ker(\varphi) \supseteq \langle xf-1\rangle$ is trivial **TODO:** . ∎

Theorem 6.8. *There is an inclusion preserving bijection*

$$\left\{\text{prime ideals of } U^{-1}R\right\} \xleftrightarrow{1:1} \{\text{prime ideals } P \subseteq R \text{ such that } P \cap U = \emptyset\}$$
$$P \mapsto \eta(P)$$

Let $Q \subseteq R$ be a prime ideal such that $Q \cap U \neq \emptyset$. Consider $\eta(Q) \cdot U^{-1}R = \langle \eta(Q)\rangle$. Let $r \in Q \cap U$, $\eta(r) \in \eta(Q) \subseteq \eta(Q) \cdot U^{-1}R$. Since $r \in U$, $\eta(r)$ is a unit in $U^{-1}R$, which implies
$$\eta(Q) \cdot U^{-1}R = U^{-1}R$$
i.e. it's not prime. ***TODO:***

Theorem 6.9.
$$\left\{\operatorname{Spec}(U^{-1}R)\right\} \xleftrightarrow{1:1} \{\mathbb{P} \in \operatorname{Spec}(R) \mid P \cap U = \emptyset\} \subseteq \operatorname{Spec}(R)$$

$$P \mapsto \eta^{-1}(P)$$

Definition 6.10. *Let R be a ring, then a basic open subset of $\operatorname{Spec}(R)$ is of the form $D(f) = \{P \in \operatorname{Spec}(R) \mid f \notin P\}$ for $f \in R$.*
And,
$$D(f) = \operatorname{Spec}(R) \setminus \mathbb{V}(I) = \operatorname{Spec}(R) \setminus \mathbb{V}(\langle f\rangle)$$
The intersection of basic open sets $D(f) \cap D(g)$ is
$$D(f) \cap D(g) = \{P \in \operatorname{Spec}(R) \mid f \notin P, g \notin P\} = \{P \in \operatorname{Spec}(R) \mid fg \notin P\} = D(fg)$$

Remark 6.11. *We know that there is a bijection,*
$$\operatorname{Spec}\left(R\left[\frac{1}{f}\right]\right) \leftrightarrow \{P \in \operatorname{Spec}(R) \mid \{1, f, f^2, \cdots\} \cap P \neq 0\} = \{P \in \operatorname{Spec}(R) \mid f \notin P\} = D(f)$$
Hence there is a bijection,
$$\operatorname{Spec}\left(R\left[\frac{1}{f}\right]\right) \leftrightarrow D(f)$$

Theorem 6.12. *Let $f \in R$ and $\eta : R \to R\left[\frac{1}{f}\right]$ be a localization map. The induced map*
$$\operatorname{Spec}\left(R\left[\frac{1}{f}\right]\right) \xrightarrow{\eta^\#} \operatorname{Spec}(R)$$
is a homeomorphism onto $D(f)$.

Definition 6.13. *The nilradical of R is the set of nilpotents*

$$\mathfrak{N}(R) = \{f \in R \mid f^k = 0, \textit{for some } k\} = \sqrt{\langle 0 \rangle}$$

Proposition 6.14. $\mathrm{Spec}(R) \cong \mathrm{Spec}(R/\mathfrak{N}(R))$ *induced by the quotinent map*

$$R \xrightarrow{\pi} R/\mathfrak{N}(R)$$

such that

$$\mathrm{Spec}(R/\mathfrak{N}(R)) \xrightarrow{\pi^{\#}} \mathrm{Spec}(R)$$

Proposition 6.15. $\mathfrak{N}(R) = \bigcap\limits_{P \in \mathrm{Spec}(R)} P$

Proof. ($\implies$)

Let $f \in \mathfrak{N}(R)$ so $f^n = 0$. Pick any prime ideal $P \subseteq R$, $f^n = 0 \in P \implies f \in P$ since P is prime.

($\impliedby$)

We show given $f \notin \mathfrak{N}(R)$ there exists a prime $P \subseteq R$ with $f \notin P$(i.e. $D(f) \neq \emptyset$). ☐

7. NONZERODIVISORS

Remark 7.1. *If M is a R-module we say an element $r \in R$ is a nonzerodivisor on $M \iff M \xrightarrow{\times r} M$ is injective $\iff r \notin Ann(m), \forall m \in M \iff (0 :_M r) = \{m \in M \mid rm = 0\} = 0$*

Definition 7.2. *Let M be a R-module. A sequence of elements $x_1, \ldots, x_t \in R$ is a **regular sequence** on M if and only if*

(1) x_i is a nonzerodivisor on $M/\langle x_1, \ldots, x_{i-1}\rangle M$
(2) $M \neq \langle x_1 \ldots x_t\rangle M$ (non-degenracy condition)

Lemma 7.3. *If $(R, \mathfrak{m})$ is a local ring, M is a finitely generated non-zero R-module, and $x_1, \ldots, x_t \in \mathfrak{m}$ then the conditions (1) and (2) above are equivalent to just (1).*

Proof. Nakayama □

Theorem 7.4 (Krull's Intersection Theorem)**.** *Let R be a Noetherian ring, $I \subseteq R$ an ideal, and M a finitely-generated R-module. There exists an element $r \in I$ such that*

$$(1 - r) \bigcap_{k=1}^{\infty} I^k M = 0$$

Corollary 7.4.1. *If $(R, \mathfrak{m})$ is a local Noetherian ring, $I \subsetneq R$ then*

$$\bigcap_{k=1}^{\infty} I^k = 0$$

Theorem 7.5. *Let $(R, \mathfrak{m})$ be a local ring, R is Noetherian, and M is a non-zero finitely generated R-module. If $x_1, \ldots, x_t \in \mathfrak{m}$ then $x_1, \ldots, x_t$ is a regular sequence on M if and only if every permutation is a regular sequence on M.*

Proof. Let x, y be a regular sequence on M. We want to show that y, x is a regular sequence on M.

Assume $ym = 0$ for some $m \in M$. Then we have $y\overline{m} = 0 \in M/xM \implies \overline{m} = 0 \implies m = xm_1$ for some m_1.

That is,

$$0 = ym = yxm_1 \implies ym_1 = 0 \in M \implies y\overline{m_1} = 0 \in M/xM$$

Then by inducting on the m_is, we get $m \in \bigcap_{k=1}^{\infty} \langle x\rangle^k M$. By KIT, this implies $m = 0$, a contradiction, and thus y is a NZD on M.

Now, assume $x\overline{m} = 0 \in M/yM$. Then, $xm = ym_1 \in M \implies y\overline{m_1} = 0 \in M/xM \implies \overline{m_1} = 0 \implies m_1 = xm_2 \in M$. So in M,

$$xm = ym_1 = xym_2 \implies m = ym_2 \implies \overline{m} = 0 \in M/yM$$

Thus x is a nonzerodivisor on M/yM, and y, x is a regular sequence. □

Definition 7.6. *Given an ideal $I \subseteq R$ the **height** of I is*

$$\begin{aligned} ht(I) &= \sup\{d \in \mathbb{Z}_{\geq 0} \mid \text{there exists a chain of prime ideals } P_0 \subsetneq P_1 \subsetneq \cdots, \subsetneq P_d \subseteq I\} \textbf{TODO:}\text{this breaks} \\ &= \dim R - \dim \mathbb{V}(I) \\ &= \dim R - \dim R/I \\ &= \sup\{ht(P) \mid P \subseteq I\} \textbf{TODO:}\text{this breaks} \\ &= \inf\{ht(P) \mid P \subseteq I\} \end{aligned}$$

Lemma 7.7. *Let R be a ring*

(1) $\dim(R) = \sup\{ht(P) \mid P \in \operatorname{Spec}(R)\}$
(2) If $(R, \mathfrak{m})$ is local, then $\dim(R) = ht(\mathfrak{m})$
(3) Let $I \subseteq R$, then $ht(I) = ht(\sqrt{I})$

Theorem 7.8. *Let R be a finitely-generated k-algebra. If R is an integral domain then for any ideal $I \subseteq R$,*

$$ht(I) = \dim(R) - \dim(R/I)$$

Definition 7.9. *Given a Noetherian local ring $(R, \mathfrak{m}, k)$, its **embedding dimension** $\mu(R)$ is the minimal number of generators required for its maximal ideal $\mathfrak{m}$.*

Theorem 7.10 (Krull's Principal Ideal Theorem)**.** *Let R be a Noetherian ring, $x \in R$. If $P \subseteq R$ is a prime ideal minimal over $\langle x \rangle$ then $ht(P) \leq 1$.*

Corollary 7.10.1. *Let R be a Noetherian ring, and ideal $I = \langle r_1, \ldots, r_n \rangle \subseteq R$. If $P \subseteq R$ is a prime ideal minimal over I then $ht(P) \leq n$.*

Theorem 7.11 (Krull's Height Theorem)**.** *If R is a Noetherian ring and $P \subseteq R$ is prime and minimal over $\langle f_1, \ldots, f_r \rangle$ (among primes, wrt inclusion). Then $ht(P) \leq r$.*

Corollary 7.11.1. *If $(R, \mathfrak{m})$ is a Noetherian local ring with residue field $k = R/\mathfrak{m}$,*

$$\dim(R) = ht(\mathfrak{m}) \leq \mu(\mathfrak{m}) = \dim_k(\mathfrak{m} / \mathfrak{m}^2)$$

Definition 7.12. *A local ring $(R, \mathfrak{m})$ is **regular** if and only if $\mathfrak{m}$ can be generated by exactly* $\dim(R)$ *many elements* $\iff \mu(\mathfrak{m}) = \dim(R)$

Definition 7.13. *Let $(R, \mathfrak{m})$ be a local ring or $\dim(R) = d$. A **system of parameters (SOP)** for R is a sequence of elements $f_1, \ldots, f_d \in \mathfrak{m}$ such that $\sqrt{\langle f_1, \ldots, f_d \rangle} = \mathfrak{m}$.*

Lemma 7.14. *Let $(R, \mathfrak{m})$ be local. A sequence $f_1, \ldots, f_d \in \mathfrak{m}$ is a SOP $\iff$ $R/\langle f_1, \ldots, f_d \rangle$ has dimension zero.*

Lemma 7.15. *Let x be a nonzerodivisor on a Noetherian ring R.*

$$\dim R/x = \dim R - 1$$

Corollary 7.15.1. *If $x_1, \ldots, x_n$ is a regular sequence on R then*

$$\dim\left(R/\langle x_1, \ldots, x_n \rangle\right) = \dim R - n$$

Definition 7.16. *Let R be a ring, $I \subseteq R$ an ideal, M a module, The depth of M with respect to I is*

$$depth_I(M) = \{\text{maximal length of a regular sequence on } M \text{ contained in } I\}$$

If $(R, \mathfrak{m})$ is local,

$$depth(M) = depth_{\mathfrak{m}}(M)$$

Example 7.1. *Depth is not always finite. For instance, if $R = M = \mathbb{K}[x_1, x_2, \ldots]$ where $x_1, \ldots, x_n$ is a regular sequence for all n.*

Proposition 7.17. *Let M be a R-module, and $x \in R$, then TFAE,*

(1) x is a regular sequence on M.
(2) $k(x, M) = 0 \to M \xrightarrow{\times x} M \to 0$
Where $H_1(k(x, M)) = 0$ and $H_0(k(x, M)) \neq 0$. In general,

$$H_1(k(x, M)) = \ker(M \xrightarrow{\times x} M) = (0 :_M x)$$

and,

$$H_0(k(x, M)) = M/xM$$

If $x, y \in R$.

$$k(x, y, M) = 0 \to M \xrightarrow{\begin{bmatrix} y \\ -x \end{bmatrix}} M \oplus M \xrightarrow{\begin{bmatrix} x & y \end{bmatrix}} M \to 0$$

In general we need,

$$H_0(k(x, y, m)) = M/(xf + yg \mid f, g \in M) = M/\langle x, y \rangle M \neq 0$$

$$H_2(k(x, y, m)) = \ker\left(\begin{bmatrix} y \\ -x \end{bmatrix}\right) = \{m \in M \mid m(-x) = 0, my = 0\}$$

Theorem 7.18 (Noether)**.** *Let R be a Noetherian ring if $I \subsetneq R$ then the set $\{\mathfrak{p} \in \operatorname{Spec}(R) \mid \mathfrak{p}$ is minimal over $I\}$ is finite.*

Lemma 7.19 (Prime Avoidance)**.** *Let R be a ring and $I, J_1, \ldots, J_n$ be proper ideals and let $J_2, \ldots, J_n$ be prime ideals. Then,*

(1) If $I \subseteq \cup_{i=1}^{n} J_i$ then $I \subseteq J_i$ for some i.
(2) If $I \not\subseteq J_i$ for all i then there exists $x \in J$ such that $x \notin \cup_{i=1}^{n} J_i$

Theorem 7.20. *Let R be a Noetherian ring and M a finitely generated R-module then*

$$\#\{\mathfrak{p} \in \operatorname{Spec}(R) \mid \mathfrak{p} = Ann(m),\ m \in M\} < \infty$$

such primes are called associated primes of M.

Lemma 7.21. *If $(R, \mathfrak{m})$ is a local Noetherian regular ring then R is an integral domain.*

Proof. We will show this by induction on the dimension of R.

Base Case: $\dim(R) = 0$. Then,

$$\dim_R(\mathfrak{m} \,/\, \mathfrak{m}^2) = \dim(R) = 0$$

Then as vector spaces $\mathfrak{m} = \mathfrak{m}^2 = \mathfrak{m}\,\mathfrak{m}$. Then Nakayama's Lemma implies that $\mathfrak{m} = \langle 0 \rangle$, and hence R is a field and thus a domain.

Inductive Step: Assume $\dim(R) = d$, and the claim is true for all rings of dim $\leq d - 1$. Let $(R, \mathfrak{m})$ be regular, i.e. $\dim(R) = \dim_R(\mathfrak{m} \,/\, \mathfrak{m}^2)$. We want to quotient R by an element $x \in R$ such that

- $R/\langle x\rangle$ has dim $= d-1$.
  By KPT $R/\langle x\rangle$ has dim $= d-1 \iff x$ is a nonzerodivisor $\iff x$ is not contained in an associated prime of R. So we must avoid the finitely many associated primes.
- $R/\langle x\rangle$ is regular.
  $R/\langle x\rangle$ is regular $\iff \dim_R(\mathfrak{n}\,/\,\mathfrak{n}^2) = \dim(R/\langle x\rangle) = d-1$, where $\mathfrak{n} = \mathfrak{m}\,/\langle x\rangle$. We claim if $x \notin \mathfrak{m}^2$ then $\dim(\mathfrak{n}\,/\,\mathfrak{n}^2) = d-1$. By prime avoidance we may pick $x \in \mathfrak{m}$ such that $x \notin \mathfrak{m}^2$ and $x \notin$ an associated prime.

Thus $(R/\langle x\rangle)$ is a regular local Noetherian ring of dimension $d-1$. So by induction it is a domain.

Now assume for contradiction that R is not a domain. Then

$$ab = 0 \quad \text{for } a, b \in \mathfrak{m}$$

reduction mod x gives us

$$\overline{a}\overline{b} = 0 \quad \text{in } R/\langle x\rangle$$

WLOG let $\overline{a} = 0 \implies a = a_1 x$. Then since x is a nonzerodivisor

$$ab = (a_1 x)b = 0 \implies a_1 b = 0$$

We can repeat this process and get,

$$a \in \cap_{n\geq 1}\langle x^n\rangle$$

which implies $a = 0$ by KIT. Thus R is a domain. □

Theorem 7.22. *If $(R, \mathfrak{m})$ is a regular local Noetherian ring then*

$$depth_{\mathfrak{m}}(R) = \dim(R)$$

Proof. We will show this by induction on the dimension of R.

Base Case: $\dim(R) = 0$. Then R is a field and $\text{depth}_{\mathfrak{m}}(R) = \dim(R)$.

Inductive Step: Assume $\dim(R) = d$, and the claim is true for all rings of dim $\leq d-1$. Pick $x \in \mathfrak{m}\setminus\mathfrak{m}^2$.Since R is a domain, x is a nonzerodivisor. Then,

$$\dim(R/\langle x\rangle) = d-1, \quad \dim(\mathfrak{n}\,/\,\mathfrak{n}^2) = d-1$$

Now pick a regular sequence $\overline{y_2}, \ldots, \overline{y_d}$ for $R/\langle x\rangle$. Pick lifts $y_2, \ldots, y_d$ in R of $\overline{y_2}, \ldots, \overline{y_d}$ look at $x, y_2, \ldots, y_d$,

- x is a nonzerodivisor
- y_2 is a nonzerodivisor in since $y_2 = \overline{y_2}$ on $R/\langle x\rangle$.

We get the rest from $\overline{y_2}, \ldots, \overline{y_d}$ being a regular sequence in $R/\langle x\rangle$. Thus $\dim(R) = \text{depth}_{\mathfrak{m}}(R)$. □

Definition 7.23. *A local ring $(R, \mathfrak{m})$ is **Cohen-Macaualy** if and only if*

$$depth(R) = \dim(R)$$

Corollary 7.23.1. *If $(R, \mathfrak{m})$ is a local Noetherian ring and R is regular then R is Cohen-Macaualy. The converse is not true.*

Definition 7.24. *Let R be a ring. An R-module is projective if and only if given any surjective R-linear mapo $f : M \twoheadrightarrow N$ and any R-linear map $g : P \to N$*

$$\begin{tikzcd} & P \arrow[d, "g"] \arrow[dl, dashed, "\widetilde{g}"'] \\ M \arrow[r, two heads, "f"] & N \end{tikzcd}$$

then there exists $\widetilde{g}$ as above.

Example 7.2. *(1) Let F be a free R-module. We claim that F is projective.*

$$\begin{tikzcd} & F \arrow[d, "g"] \arrow[dl, dashed, "\widetilde{g}"'] \\ M \arrow[r, two heads, "f"] & N \end{tikzcd}$$

g is determined on a basis $\{e_i\}$. Then we define $\widetilde{g}$ on lifts of $g(e_i)$.

(2) Let $R = \mathbb{Z}/6\mathbb{Z}$ and $M = \langle 3 \rangle = \{\overline{0}, \overline{3}\}$. Then M is not free since free modules of R have either 1 element, 6^n elements, or infinite elements. We have $R \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$. And $M \cong \mathbb{Z}/2\mathbb{Z}$. So M is a summand of a free-module. Then by the theorem below, M is projective, not free.

Theorem 7.25. *Let R be a ring and P an R-module. Then TFAE,*

(1) P is projective.
(2) Every surjection $M \twoheadrightarrow P$ splits.
(3) P is a direct summand of a free module.
(4) the functor $\mathrm{Hom}_R(P, -)$ is exact.

Proof. (1) $\implies$ (2)

Pick a surjection. Then,

$$\begin{tikzcd} & P \arrow[d, "\mathrm{id}"] \arrow[dl, dashed] \\ M \arrow[r, two heads] & P \end{tikzcd}$$

(2) $\implies$ (3)

Pick a surjection with F free

$$0 \to k \to F \twoheadrightarrow P \to 0$$

by (2) this splits on right and by splitting lemma $F \cong P \oplus k$.

(1) $\equiv$ (4)

Since $\mathrm{Hom}_R(P, -)$ is injective.

(3) $\implies$ (1)

Assume $F \cong P \oplus k$ with F free.

$$\begin{tikzcd}
 & P \arrow[d, "g"] \\
M \arrow[r, two heads, "f"] & N
\end{tikzcd}$$

Direct Sum-ing with k,

$$\begin{tikzcd}
 & P \oplus k \cong F \arrow[d, "g\oplus\mathrm{id}"] \\
M \oplus k \arrow[r, two heads, "f\oplus\mathrm{id}"] & N \oplus k
\end{tikzcd}$$

TODO: Then there exists a map $F \to M \oplus k$

Hence proved. □

Definition 7.26. *A matrix $E \in Mat_{n\times n}(R)$ is **idempotent** if and only if $E^2 = E$.*

Remark 7.27. *If E is an idempotent then $I - E$ is an idempotent.*

Lemma 7.28. *Let R be a finitely generated R-module then P is projective if and only if there exists an idempotent matrix $E \in Mat_{n\times n}(R)$ for some $n > 0$ such that $P \cong \mathrm{im}(R^n \xrightarrow{E} R^n)$.*

Proof. ($\Longrightarrow$)

Assume P is projective. Since P is finitely generated

$$0 \to k \to R^n \to P \to 0$$

Since $R^n \to P$ splits $R^n \cong P \oplus k$. Now let $E : R^n \to R^n$ be a projection onto P. (Projection is idempotent).

($\Longleftarrow$)

Assume $P \cong \mathrm{im}(R^n \xrightarrow{E} R^n)$ with $E^2 = E$. Then,

$$R^n \cong \mathrm{im}(E) \oplus \ker(E)$$

and $\mathrm{im}(E) \cong P$. □

Definition 7.29. *A module M is **finitely-presented** if there exists an exact sequence*

$$R^m \to R^n \twoheadrightarrow M \to 0$$

for some m, n.

Remark 7.30. *If M is finitely-generated*

$$\begin{tikzcd}
0 \arrow[r] & R^m \arrow[r] & R^n \arrow[r, two heads] & M \arrow[r] & 0 \\
 & R^m \arrow[u, two heads] \arrow[ur, dashed] & & &
\end{tikzcd}$$

Being finitely-presented is the same as being finitely-generated and the kernel of the surjective map being fg. ***TODO:***

Theorem 7.31. *Let R be a ring and let M be a finitely-presented R-module. The following are equivalent:*

(1) *M is projective*
(2) *$M_{\mathfrak{p}}$ is a free $R_{\mathfrak{p}}$ module for all $\mathfrak{p} \in \operatorname{Spec}(R)$*
(3) *$M_{\mathfrak{p}}$ is free as an $R_{\mathfrak{p}}$ module for all $\mathfrak{p} \in \operatorname{mSpec}(R)$*
(4) *$M[1/f_i]$ is free for $i = 1, \ldots, n$ where $\langle f_1, \ldots, f_t \rangle$*

Theorem 7.32. *Let $(R, \mathfrak{m})$ be a local ring. If P is a finitely-presented projective R-module then P is free.*

Proof. Let $(R, \mathfrak{m}, k)$ be a local ring where k is the residue field and let P be a finitely-presented R module that is projective. Pick $y_1, \ldots, y_t \in P$ whose images are a basis for $P/\mathfrak{m}P$ as a k-vector space (Nakayama). Then we have a surjection,

$$R^t \twoheadrightarrow P, \quad e_i \mapsto y_i$$

Since P is projective this map splits,

$$R^t \cong P \oplus k$$

for a finitely-generated module k. This gives us a map

$$R^n \twoheadrightarrow P \oplus k, \quad e_i \mapsto (y_i, 0)$$

Tensoring by R/fm gives us

$$k^n \twoheadrightarrow P/\mathfrak{m}P \oplus k/\mathfrak{m}k, \quad e_i \mapsto \overline{y_i}$$

Which implies $k/\mathfrak{m}k = 0$. By Nakayama's lemma applied to k,

$$\implies k = 0 \implies R^t \cong P$$

□

Theorem 7.33 (Quillen-Suslin)**.** *If k is a field and M is a finitely-presented projective module over $k[x_1, \ldots, x_n]$ then M is free.*

Geometrically speaking this means "over $\mathbb{A}^n$ every algebraic" vector bundle of finite rank is trivial.

Definition 7.34. *Let R be a ring and M be a R-module. A **projective resolution** of M is a chain complex*

$$0 \leftarrow P_0 \xleftarrow{d_0} P_1 \xleftarrow{d_1} P_2 \xleftarrow{d_2} \cdots$$

such that

(1) *The P_i are projective for all i.*
(2) $H_i(P_\bullet, d_\bullet) = \begin{cases} M & i = 0 \\ 0 & \textit{otherwise} \end{cases}$

*We say that this is a **free resolution** if P_i is free for all i.*

Definition 7.35. *The **projective dimension** of a module M is*

$$pd(M) = \inf\{\textit{lengths of projective resolutions of } M\}$$

Definition 7.36. *A free resolution $F_\bullet$ of a module M is **minimal** if and only if for each differential*

$$F_i \xleftarrow{d_i} F_{i+1}$$

we have that $\operatorname{img}(d_i) \subseteq \mathfrak{m}\, F_i$

Definition 7.37. *If $F_\bullet$ and $G_\bullet$ are chain complexes of R-modules, then an isomorphism $\psi_\bullet : F_\bullet \to G_\bullet$ is a collection $\{\psi_i : F_i \to G_i\}$ of R-module maps such that*

(1) ψ_i is an isomorphism for all i and,

(2)
$$\begin{tikzcd} F_i \arrow[d, "\psi_i"] & F_{i+1} \arrow[l, "d_i"'] \arrow[d, "\psi_{i+1}"] \\ G_i & G_{i+1} \arrow[l, "\delta_i"'] \end{tikzcd}$$

Theorem 7.38. *Let $(R, \mathfrak{m})$ be a Noetherian local ring and M a finitely-generated R-module, then*

(1) A minimal free resolution of M exists
(2) If $F_\bullet$ and $G_\bullet$ are minimal free resolutions of M then $F_\bullet \cong G_\bullet$

Proof. (1) Pick a minimal generating set of M $m_1, \ldots, m_t$ such that

$$0 \leftarrow M \overset{\pi}{\twoheadleftarrow} F_\bullet \cong R^{\oplus t} \leftarrow \ker(\pi) \leftarrow 0$$

Since R is Noetherian and $F_\bullet$ is finitely-generated, $\ker(\pi)$ is finitely-generated. Pick a minimal generating set $k_1, \ldots, k_s$ such that

$$\ker(\pi) \overset{\pi_1}{\twoheadleftarrow} F_1 \cong R^{\oplus S}, \quad k_i \leftarrow\!\shortmid e_i$$

So we have,

$$\begin{tikzcd} & & & 0 & \\ 0 & M \arrow[l] & F_0 \arrow[l, two heads, "\pi"'] & \ker(\pi) \arrow[l] \arrow[u] & 0 \arrow[l] \\ & & & F_i \arrow[u, two heads, "\pi_1"] \arrow[ul, "d_0"] & \\ & & & \ker(\pi_2) \arrow[u] & \\ & & & 0 \arrow[u] & \end{tikzcd}$$

such that $\operatorname{coker}(d_0) \cong M$ and $\operatorname{img}(d_0) \cong \mathfrak{m}\, F_0$ since none of $m_1, \ldots, m_t$ lie in $\mathfrak{m}\, M$. The rest of the proof follows by induction. $\square$

Theorem 7.39 (Kaplansky)**.** *Let $(R, \mathfrak{m})$ be a local ring and P is any R-module then P is projective if and only if P is free.*

Definition 7.40. *Let $F_\bullet$ and $G_\bullet$ be projective chain complexes* ***a chain homotopy*** *between chain maps $\psi_\bullet$ and $\phi_\bullet$ is the data of R-module maps $h_i : F_i \to G_{i+1}$ such that,*

$$\phi_{i+1}(x) - \phi_{i+1}(x) = h_i(d_i(x)) + \delta_{i+1}(h_{i+1}(x))$$

$$\begin{tikzcd}
F_i \arrow[d, shift right, "\psi_i"'] \arrow[d, shift left, "\phi_i"] \arrow[dr, "h_i"] & F_{i+1} \arrow[l, "d_i"'] \arrow[d, shift right, "\psi_{i+1}"'] \arrow[d, shift left, "\phi_{i+1}"] \arrow[dr, "h_{i+1}"] & F_{i+2} \arrow[l, "d_{i+1}"'] \arrow[d, shift right, "\psi_{i+2}"'] \arrow[d, shift left, "\phi_{i+2}"] \\
G_i & G_{i+1} \arrow[l, "\delta_i"'] & G_{i+2} \arrow[l, "\delta_{i+1}"']
\end{tikzcd}$$

Theorem 7.41. *Let R be a ring, M and N R-modules and*

$$0 \leftarrow M \xleftarrow{\pi_M} F_\bullet, \quad 0 \leftarrow N \xleftarrow{\pi_N} G_\bullet$$

be projective resolutions of M and N respectively. Given any R-module map $\phi : M \to N$ there exists a lift f_0 of ϕ to $F_\bullet$ and $G_\bullet$ i.e.

$$\begin{tikzcd}
0 & M \arrow[l] \arrow[d, "\phi"] & F_0 \arrow[l, "\pi_M"'] \arrow[d, "f_0"] & F_1 \arrow[l, "d_0"'] \arrow[d, "f_1"] & F_2 \arrow[l, "d_1"'] \arrow[d, "f_2"] & \cdots \arrow[l] \\
0 & N \arrow[l] & G_0 \arrow[l, "\pi_N"'] & G_1 \arrow[l, "\delta_0"'] & G_2 \arrow[l, "\delta_1"'] & \cdots \arrow[l]
\end{tikzcd}$$

Moreover the lift f_0 is unique upto homotopy.

Corollary 7.41.1. *Projective residue are unique upto chain homotopy.*

Corollary 7.41.2. *Say $(R, \mathfrak{m})$ is a local Noetherian ring and $F_\bullet$ and $G_\bullet$ are minimal free resolutions of a finitely-generated module M then*

(1) $F_\bullet$ and $G_\bullet$ are homotopy equivalent.
(2) If $\phi_\bullet, \psi_\bullet : F_\bullet \to G_\bullet$ are chain maps that are homotopy equivalent then

$$\psi_\bullet \otimes R/\,\mathfrak{m} = \phi_\bullet \otimes R/\,\mathfrak{m}$$

Proof. Suppose h_i is a homotopy then

$$\phi_{i+1}(x) - \phi_{i+1}(x) = h_i(d_i(x)) + \delta_{i+1}(h_{i+1}(x))$$

the minimality of $F_\bullet$ implies $\operatorname{img}(d_i) \subseteq \mathfrak{m}\, F_{i+1}$ and the minimality of $G_\bullet$ implies $\operatorname{img}(\delta_i) \subseteq \mathfrak{m}\, G_{i+1}$. That is $\operatorname{img}(\psi_{i+1} - \phi_{i+1}) \subseteq \mathfrak{m}\, G_{i+1}$. Tensoring by $R/\,\mathfrak{m}$ gives

$$F_{i+1}/\,\mathfrak{m}\, F_{i+1} \xrightarrow{\overline{\psi_{i+1}} - \overline{\phi_{i+1}}} G_{i+1}/\,\mathfrak{m}\, G_{i+1}$$

is the zero map which implies that

$$\psi_{i+1} \otimes R/\mathfrak{m} = \phi_{i+1} \otimes R/\mathfrak{m}$$

□

Definition 7.42. *The exterior product $\Lambda^{i+1}(R^n)$ is the free R-module with symbols $e_{j_1} \wedge e_{j_2} \wedge \cdots \wedge e_{j_{i+1}}$ modulo the alternating relations.*

Definition 7.43. *Let R be a ring and $x_1, \ldots, x_n$ be a sequence of elements in R. The **Koszul complex** on $x_1, \ldots, x_n$ is the chain complex of R-module $K_\bullet^R(x_1, \ldots, x_n)$ where*

$$K_{i+1}^R(x_1, \ldots, x_n) = \Lambda^{i+1}(R^n) \xrightarrow{d_i} \Lambda^i(R^n) = K_i^R(x_1, \ldots, x_n)$$

where

$$d_i(e_{j_1} \wedge e_{j_2} \wedge \cdots \wedge e_{j_{i+1}}) = \sum_{k=1}^{i+1} (-1)^k x_{j_k} \cdot e_{j_1} \wedge \cdots \wedge \widehat{e_{j_k}} \wedge \cdots \wedge e_{j_{i+1}}$$

Proposition 7.44. *$K_\bullet^R(x_1, \ldots, x_n)$ is a free chain complex.*

Proof. We check $\delta_{i-1}(\delta_i(-)) = 0$ on basis element. And

$$\begin{aligned}
\delta_{i-1}(\delta_i(e_{j_1} \wedge e_{j_2} \wedge \cdots \wedge e_{j_{i+1}})) &= \delta\left(\sum_{k=1}^{i+1} (-1)^k x_{j_k} \cdot e_{j_1} \wedge \cdots \wedge \widehat{e_{j_k}} \wedge \cdots \wedge e_{j_{i+1}}\right) \\
&= \sum_{k=1}^{i+1} (-1)^k x_{j_k} \cdot \delta(e_{j_1} \wedge \cdots \wedge \widehat{e_{j_k}} \wedge \cdots \wedge e_{j_{i+1}}) \\
&= \sum_{k=1}^{i+1} (-1)^k x_{j_k} \cdot \left(\sum_{\ell=1}^{i+1} (-1)^{\ell-1} x_{j_\ell} \cdot (e_{j_1} \wedge \cdots \wedge \widehat{e_{j_k}} \wedge \cdots \widehat{e_{j_\ell}} \wedge \cdots \wedge e_{j_{i+1}})\right)
\end{aligned}$$

TODO:

Then this is the zero map. □

Lemma 7.45. *$K_\bullet^R(x_1)$ is a free resolution $R/\langle x_1 \rangle$ if and only if x_1 is a nonzerodivisor if and only if x_1 is a regular sequence. Then the complex*

$$0 \to R \xrightarrow{-x_1} R \to R/\langle x_1 \rangle \to 0$$

upon tensoring by M

$$0 \to M \xrightarrow{-x_1} M \to M/\langle x_1 \rangle M \to 0$$

remains exact.

Definition 7.46. *Let R be a ring, M an R-module and $x_1, \ldots, x_n$ a sequence of elements in R. The **koszul complex** of $x_1, \ldots, x_n$ on M*

$$K_\bullet^R(x_1, \ldots, x_n) = K_\bullet^R(\overline{x}, M)$$

*The **koszul homology** of $x_1, \ldots, x_n$ on M is $H_i(\overline{x}, M) = H_i(K_\bullet^R(\overline{x}, M))$.*

Theorem 7.47. *Let R be a ring and M be a R-module. If $x_1, \dots, x_n$ is a regular sequence on M then*

$$H_i(\overline{x}, M) = \begin{cases} 0 & i > 0 \\ M/\langle x_1, \dots x_n \rangle M & i = 0 \end{cases}$$

Corollary 7.47.1. *If $x_1, \dots, x_n$ is a regular sequence on R then the koszul complex of $K^R_\bullet(x_1, \dots, x_n)$ is a free resolution of $R/\langle x_1, \dots, x_n \rangle$.*

Theorem 7.48. *Let $(R, \mathfrak{m})$ be a local ring and M a finitely-generated R-module. Let $x_1, \dots, x_n \in \mathfrak{m}$ then $x_1, \dots, x_n$ is a regular sequence on M if and only if $H_i(\overline{x}, M) = \begin{cases} 0 & i > 0 \\ M/\langle x_1, \dots x_n \rangle M & i = 0 \end{cases}$*

Definition 7.49. *Let $(R, \mathfrak{m})$ be a local ring, $k = R/\mathfrak{m}$ be it's residue field, and M be a R-module.*

$$Tor^i_R(k, M) = H_i(F_\bullet \otimes M)$$

where $F_\bullet$ is the free resolution of k as an R-module. And

$$Ext^i_R(k, M) = H_i(\mathrm{Hom}_R(F_\bullet, M))$$

Remark 7.50 (Tor is Balanced). *If $G_\bullet$ is a free resolution of M then*

$$Tor^i_R(M, k) = H_i(G_\bullet \otimes k) \cong H_i(F_\bullet \otimes M) \cong Tor^i_R(k, M)$$

Lemma 7.51. *Let $H_\bullet$ be a free resolution of M and $G_\bullet$ be a minimal free resolution of M then*

$$H_\bullet \cong G_\bullet \oplus C_\bullet$$

That is, $pd(M) =$ length of the minimal free resolution.

Lemma 7.52. *Let M be a finitely-generated module over a Noetherian local ring $(R, \mathfrak{m})$ with residue field k. Then*

$$pd_R(M) = \sup\{i \in \mathbb{Z}_{\geq 0} \mid Tor^i_R(M, k) \neq 0\}$$

Proof. $\sup\{i \in \mathbb{Z}_{\geq 0} \mid \mathrm{Tor}^i_R(M, k) \neq 0\} \leq \mathrm{pd}_R(M)$. Compute $\mathrm{Tor}(k, M)$ by the minimal free resolution of M.

$$\mathrm{Tor}^i(M, k) = H_i(G_\bullet \otimes R) = H_i(G_\bullet \otimes R/\mathfrak{m}) = \dim_R(G_i/\mathfrak{m}\, G_i)$$

TODO: □

Theorem 7.53 (Auslander-Buchsbaum Formula). *Let $(R, \mathfrak{m}, k)$ be a local Noetherian ring, and let M be a finitely-generated R-module. If $pd_R M < \infty$ then,*

$$pd_R(M) + depth_\mathfrak{m}(M) = depth_\mathfrak{m}(R)$$

Definition 7.54. *The standard $\mathbb{Z}$-grading on $S = k[x_0, \dots, x_n]$ is a decomposition of S as an abelian group*

$$S \cong \bigoplus_{d \in \mathbb{Z}} S_d$$

where S_d is the k-vector space spanned by monomials of degree k.

A graded module over S is an S-module with a decomposition as an abelian group

$$M \cong \bigoplus_{d \in \mathbb{Z}} M_d$$

such that $S_e \cdot M_d \subseteq M_{e+d}$.

Theorem 7.55 (Hilbert's Syzygy Theorem)**.** *If M is a finitely-generated graded module over S then M has a free resolution of length at most $(n+1)$.*

Theorem 7.56 (Eisenbud)**.** *Let $(S, \mathfrak{m})$ be a regular local Noetherian ring and $f \in \mathfrak{m}^2$, $f \neq 0$ and is not a zero-divisor. If M is a finitely-generated module over $R = S/\langle f \rangle$ then the minimal R-free resolution of M is eventually 2-periodic*

$$0 \leftarrow M \leftarrow F_0 \leftarrow \cdots \leftarrow F_t \leftarrow R^s \xleftarrow{A} R^s \xleftarrow{B} R^s \xleftarrow{A} R^s \xleftarrow{B} \cdots$$

for some matrices A and B such that

$$AB = BA = f\,\mathrm{Id}$$

Theorem 7.57 (? i think this is a definition lol)**.** *Let $(R, \mathfrak{m}, k)$ be a local Noetherian ring and M a finitely-generated R-module. The poincare series of M*

$$P_M(t) = \sum_{i \geq 0} \dim_R Tor_i^R(M, R) t^i$$

where $\dim Tor_i^R(k,k) = \binom{n}{i}$ *and*

$$P_R(t) = \sum_{i \geq 0} \binom{n}{i} t^i = (1+t)^n$$

Theorem 7.58 (Auslander-Buchsbaum-Serie)**.** *Let $(R, \mathfrak{m})$ be a Noetherian local ring with residue field $k = R/\mathfrak{m}$. Then TFAE,*

(1) R is a regular ring.
(2) The global dimension of R is finite.
(3) Every finitely-generated R-module has a projective resolution of finite length.
(4) k has a projective resolution of length equal to $\dim(R)$.

Theorem 7.59 (Auslander-Buchsbaum-Serie 2)**.** *Let $(R, \mathfrak{m})$ be a Noetherian local ring with residue field $k = R/\mathfrak{m}$. Then TFAE,*

(1) R is a regular ring.
(2) The global dimension of R is finite.
(3) Every finitely-generated R-module has a projective resolution of finite length.
(4) $P_R(t)$ is a polynomial.